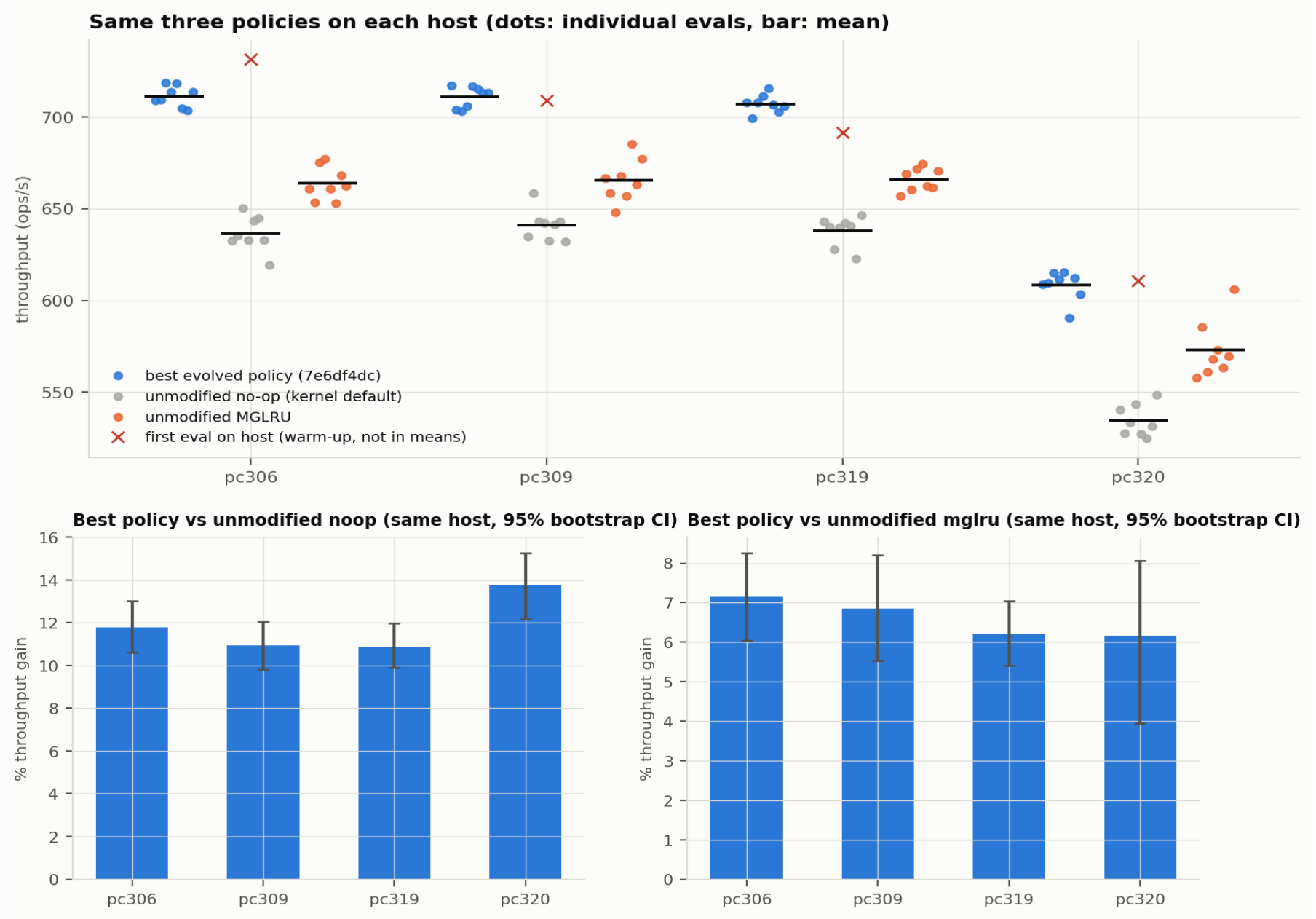
Does the host change the best policy's performance?

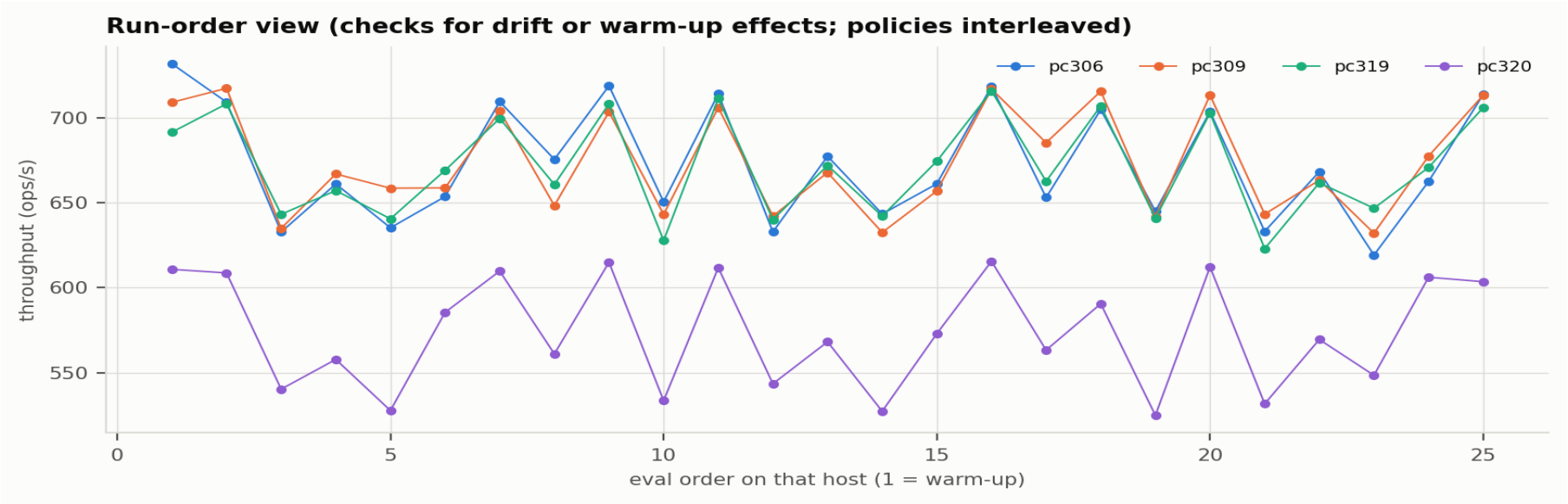
The best policy from the 4-host run (7e6df4dc, region-CLOCK, 712.2 ops/s on pc309) was run unchanged on all four hosts, interleaved with two unmodified controls (the no-op kernel default and MGLRU) so that host speed can be separated from policy effect. 8 evaluations per policy per host, order rotated each round, one warm-up eval per host (not in the means). Hosts ran in parallel, one eval at a time each.

host	best policy mean ± CI95 (sd, cv)	no-op mean ± CI95	MGLRU mean ± CI95	warm-up eval	speed index (no-op)	gain vs no-op [95% CI]	gain vs MGLRU [95% CI]
pc306	711.4 ± 4.8 (sd 5.7, cv 0.8%)	636.4 ± 8.1	664.0 ± 7.6	731.6	1.039	+11.8% [+10.6, +13.0]	+7.1% [+6.0, +8.3]
pc309	711.1 ± 4.9 (sd 5.8, cv 0.8%)	640.9 ± 7.1	665.5 ± 9.8	709.0	1.046	+11.0% [+9.8, +12.0]	+6.9% [+5.5, +8.2]
pc319	707.2 ± 4.1 (sd 4.9, cv 0.7%)	637.9 ± 6.9	666.0 ± 5.2	691.4	1.042	+10.9% [+9.9, +12.0]	+6.2% [+5.4, +7.0]
pc320	608.3 ± 6.8 (sd 8.1, cv 1.3%)	534.6 ± 7.1	573.0 ± 13.2	610.7	0.873	+13.8% [+12.2, +15.2]	+6.2% [+4.0, +8.1]

Do hosts differ in how much the policy helps? Pairwise difference in the best-vs-no-op gain (log-ratio, % points) with 95% bootstrap CI: pc306 vs pc309: +0.7 [-0.7, +2.3]; pc306 vs pc319: +0.8 [-0.7, +2.3]; pc306 vs pc320: -1.8 [-3.5, -0.0] *; pc309 vs pc319: +0.1 [-1.3, +1.4]; pc309 vs pc320: -2.5 [-4.1, -0.8] *; pc319 vs pc320: -2.6 [-4.1, -0.9] *. * = CI excludes 0.

Drift over the experiment (change in normalised throughput from first to last eval, linear fit): pc306 -0.5%, pc309 +0.6%, pc319 +0.1%, pc320 +1.1%.


Same three policies on each host (dots: individual evals, bar: mean)
throughput (ops/s)
700
650
600
550
best evolved policy (7e6df4dc)
unmodified no-op (kernel default)
unmodified MGLRU
first eval on host (warm-up, not in means)
pc306
pc309
pc319
pc320
Best policy vs unmodified noop (same host, 95% bootstrap CI)
% throughput gain
16
14
12
10
8
6
4
2
0
pc306
pc309
pc319
pc320
Best policy vs unmodified mglru (same host, 95% bootstrap CI)
% throughput gain
8
7
6
5
4
3
2
1
0
pc306
pc309
pc319
pc320



Per host and policy

host	policy	n	mean	sd	min	max	MiB read (mean)	refaults (mean)
pc306	best	8	711.4	5.7	703.4	718.7	197	24625
pc306	noop	8	636.4	9.7	619.1	650.4	182	21044
pc306	mglru	8	664.0	9.1	653.0	677.2	195	24310
pc309	best	8	711.1	5.8	703.3	717.2	191	23278
pc309	noop	8	640.9	8.5	632.0	658.5	175	19578
pc309	mglru	8	665.5	11.7	648.2	685.2	189	23129
pc319	best	8	707.2	4.9	699.5	715.6	194	23948
pc319	noop	8	637.9	8.2	622.9	646.7	184	21402
pc319	mglru	8	666.0	6.3	657.0	674.3	193	23976
pc320	best	8	608.3	8.1	590.5	615.5	169	18300
pc320	noop	8	534.6	8.5	524.9	548.4	162	16693
pc320	mglru	8	573.0	15.8	557.8	606.1	167	18131